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RESEARCH-ARTICLE

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A REFERENCE ARCHITECTURE FOR MILITARY SIMULATIONS WITH EXPERIMENTATION AND ANALYSIS FEATURES

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ABSTRACT

For the last few decades, there has been an increasing effort to incorporate simulations for decision support purposes in the military domain. Although the model design, scenario development, and execution phases are well studied, a generalized view of the overall process including experiment design, analysis, and evaluation phases are usually handled either by assuming hand-crafted approaches or via existing third-party tools. This study proposes a reference architecture, not only for developers to cite but also for practitioners to conceptualize their experimentation process. Moreover, we argue that multi-criteria decision making techniques can be used in the analysis phase to compare alternative simulation scenario outputs.

Keywords: effectiveness evaluation, reference architecture, decision support.

1 INTRODUCTION

Simulations are widely used for analysis purposes in the military domain for various problems such as procurement decisions, tactical evaluations, decision support for modernization and design. In order to achieve meaningful results, they require thorough experimentation and comparison of the alternative outputs in terms of their operational effectiveness as well as cost effectiveness. However, most of the current literature on military analysis, utilizing simulation as an analysis tool, focus on one measure of effectiveness to express performance of a system or a scenario (Lei et al. 2017, Seo et al. 2011, Park et al. 2010, Nguyen 2014, Cheng et al. 2015). The comparison of alternatives in terms of their effectiveness requires comparing conflicting measures of performance (MoP) which constitute the effectiveness of a system. To handle this problem, utilization of Multi-Criteria Decision Making (MCDM) methods is common (Hootman and Whitcomb 2005, Jiao et al. 2013, Na and Karr 2016).

This study proposes a reference architecture to aid building up an end-to-end simulation system, which allows to design, execute and analyze a military scenario. Moreover, we argue that it is possible in the analysis

Figure 1: A sub-system and data flow view of the proposed reference architecture.

2 REFERENCE ARCHITECTURE

2.1 Pre-Experiment Phase

Scenario Design During Scenario Design, the user is expected to define scenario parameters such as the location, starting time, conditional branching in time, unexpected events etc. Initial events and preferred courses of action -if any- for each model are provided during scenario definition. It should be noted that some scenario parameters can also be defined as stochastic parameters and/or design factors. Thus, scenario definitions are almost always dependent on the model definitions at the data level.

2.2 Experimentation Phase

Experiment Design In the Experiment Design stage, design factors which will be used in the specific experiment is determined with their respective levels. Multiple experimental designs can be constructed in order to analyze different aspects of a given scenario. Experimental design methods such as full factorial designs or D-optimal designs can be used considering the trade-off between computational resources and analysis requirements (Kelton 1999, Kleijnen et al. 2005). Stochastic parameters and their respective representative distributions are also defined for a given experiment. Number of trials for each experimental design point is determined after experiment specific parameters are selected. In order to provide variability among trials, random number generation (RNG) techniques are to be used to calculate actualization of stochastic parameters (L'Ecuyer 2017). All of the design settings, which are naturally dependent on scenarios, can be persisted for re-use.

Simulation Execution A managing process (likely attached with the experimental design) calls the scenario execution program which interprets and executes scenario variants generated for separate levels of experimental design factors. This capability can be implemented in a number of ways. A typical solution would be to sequentially call the execution program for each point (scenario instance) with a predefined random seed at the very beginning of multiple trial sets. Alternatively one can parallelize, or even distribute the executions, for which a RNG initialization phase may be required to manage variations emerging from random variables. Typically, scenario execution phase produces (either during, or at the end of an episode or experiment) simulation outputs for further processing at the following phases. Simulation outputs can be categorized into three: (1) developer logs for debug purposes, (2) simulation logs for future analysis and debriefing (3) MoP logs for decision support. Remember that simulation outputs should be attached to the related design points via a predefined data management setting, like directory structures (file system), relations (relational DBMS), etc.

2.3 Post-Experiment Phase

Reporting and Debriefing Formatted display and statistical methods are applied on simulation outputs at this stage. Reported data can include, but is not limited to, descriptive statistics such as means, variances, quartiles and confidence intervals of performance measures (Law 2013). The effects of design factors and relationships between them are analyzed to compare the possible design alternatives. Queries and visualization methods can also be utilized to analyze results. If the desired statistical accuracy of the results is not obtained, additional trials are defined by going back to experimental design stage (Montgomery 2017).

Analysis and Decision Support The proposed architecture also includes a decision support stage to deal with multiple conflicting objectives which need to be optimized simultaneously. Although other MCDM methods can be used to overcome this problem, we suggest the usage of *effectiveness trees* or *value trees* (Keeney 1996, Dong et al. 2008, Chengxi Dong et al. 2008) as a more general tool. Effectiveness trees enable users to utilize several a priori MCDM methods as well as other aggregation techniques at once to compare different alternatives which have several performance criteria that can be represented hierarchically. The outputs of Decision Support stage can also be used in reporting and debriefing stage for statistical analysis and reporting purposes.

3 CASE STUDY: SHIP COUNTERMEASURE SYSTEM EFFECTIVENESS

The proposed reference architecture is implemented for a small scale anti-ship missile defence scenario. In this scenario, we aim to analyze the soft-kill effectiveness of chaff usage against an RF seeker guided anti-ship missile (ASM). A ship model carrying a radar system, a chaff dispenser, and predefined number

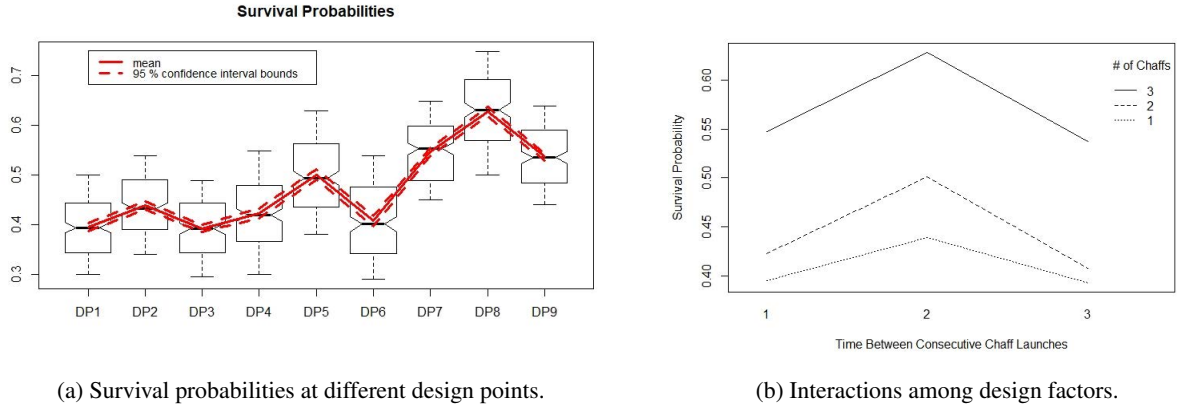
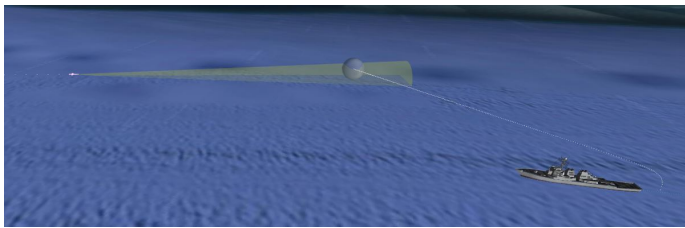


Figure 2: Sample output plots for experiment reporting.

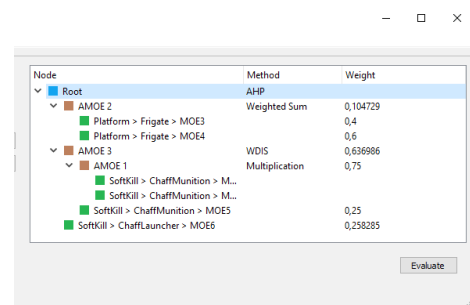
of chaffs in the dispenser are defined. An ASM is defined as the threat in the scenario. In our design, all models are agent based, interacting with their respective environments (such as the RF environment and –what we call– the Newtonian physics environment). All the data required for further analysis are to be stored (including the positions, time stamps, radar detections etc.). The parameters related to chaff fire pattern algorithm are selected as the candidates for design factors to be evaluated in the experimental design. Motor parameters of ASM are determined as the possible stochastic parameters in order to make missile movement vary stochastically in each trial.

A scenario mainly consists of a reference location and time, among other information, like initial positions and velocities of the ship and ASM. For simplicity, both radar and seeker of the ASM are assumed active upon initialization. Search, lock-on and fuse activation behaviour lists are defined for the ASM. The ship is initially defined to move according to the deployment behaviour list. When ship's radar detects the ASM, a self-defence behaviour list is activated in which the ship initiates its chaff countermeasure system according to the dispenser logic defined by design parameters.

Number of chaffs fired and the time between consecutive chaff dispenses are determined as the design factors for this specific experiment. Three levels for each factor is determined and a full factorial design is constructed. Initial number of trials for all design points is determined. Motor thrust of anti-ship missile at a given point in time is set as a random variate and its representative distribution is determined. Finally, a seed is provided for RNG. Sample statistical analysis for survival probability of the ship is performed and an example analysis is provided in Figures 2a and 2b.



(a) Visualization using SIMDIS SDK.



(b) Effectiveness tree.

Figure 3: Post processing of an experiment: Debriefing and analysis.

Figure 2a represents box plot data as well as the confidence intervals for the mean survival probability for each design point in the designed experiment. Figure 2b shows the effects of interactions between different levels of design factors on survival probability. One of the trials is visualized with the help of SIMDIS SDK (U.S. Naval Research Laboratory 2019). A sample effectiveness tree is provided for mock-up measures of effectiveness (MOE) in Figure 3b. For the given scenario the anonymous MOEs include measures related to ship, chaff dispenser and chaff munitions. In the effectiveness tree provided in Figure 3b, in order to calculate aggregate measures of effectiveness, AMOE 1, AMOE 2, AMOE 3, and overall effectiveness value, root, multiplication, weighted sum, weighted distance to ideal point and Analytic Hierarchy Process (AHP) are used respectively.

Our implementation obeys the partitioning, flows and dependencies proposed in Figure 1. As far as we have experienced, the featured benefits of this reference architecture are *reusability* and *scalability*. From the developers' point of view, each stage is by itself a software project and can be managed separately. Separation of simulation execution provides an isolated opportunity to scale up via parallelization and distribution. From the users' point of view, the control flow and data dependencies are well defined, and understandable. Moreover, depending on the design, user roles are easier to assign and administrate, even at the data level.

4 CONCLUSION

This study aims to propose a comprehensive, yet simple, reference architecture for designing military simulation software which involves experiment design, batch runs and analysis-evaluation phases. An implementation of the architecture is summarized as a case study, and the experiences are briefly discussed.

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